



# Analytics & Big Data

Session 10: Analytics in organizations

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(2026-05-11)

## Learning objectives

- **Explain** the strategic role of data analytics in data-driven organizations.
- **Discuss** the ethical and legal boundaries of data analytics in organizations.

# Industry-level perspectives

# Does the deployment of analytics pay off?

Müller et al. (2018): The Effect of Big Data and Analytics on Firm Performance

The study examines whether firms with live **big data and analytics (BDA) assets** show higher productivity than comparable firms without such assets.

## Empirical setting

- 814 publicly traded firms
- 2008–2014 panel data
- 5,698 firm-year observations
- Objective data on live BDA assets
- Financial data from Compustat

## Outcome and controls

- Outcome:  $\log(\text{Sales})$
- Inputs: labor, capital, materials
- IT controls: ERP, CRM, SCM
- Industry and year controls
- Industry moderators: IT intensity and competition

$$\log(\text{Sales}) = \beta_0 + \beta_1 \log(\text{Labor}) + \beta_2 \log(\text{Capital}) + \beta_3 \log(\text{Materials}) + \beta_4 \text{BDA} + \text{Controls} + \varepsilon$$



|                   | OLS                 | FE                  |
|-------------------|---------------------|---------------------|
| log(Capital)      | 0.090***<br>(0.015) | 0.127***<br>(0.029) |
| log(Labor)        | 0.296***<br>(0.024) | 0.472***<br>(0.043) |
| log(Materials)    | 0.667***<br>(0.023) | 0.442***<br>(0.035) |
| BDA               | 0.041**<br>(0.020)  | 0.038*<br>(0.021)   |
| Industry dummies? | Yes                 | Yes                 |
| Year dummies?     | Yes                 | Yes                 |
| IT Asset dummies? | Yes                 | Yes                 |
| Observations      | 5,698               | 5,698               |
| $R^2$             | 0.977               | 0.791               |
| Adjusted $R^2$    | 0.977               | 0.755               |

## Key findings and implications

- **Analytics assets alone are insufficient:** value depends on complementary resources such as data infrastructure, enterprise systems, and analytical skills.
- **Industry conditions matter:** IT-intensive and highly competitive industries show clearer productivity effects.
- **Causality remains difficult:** observational firm-level data can reduce, but not fully eliminate, concerns about reverse causality and omitted variables.

At the industry level, analytics deployment should be understood as part of a broader configuration of digital capabilities, competitive pressure, and organizational capacity to turn data into decisions.

# A meta-analysis



Oesterreich et al. (2022): What translates big data into business value?

This meta-analysis examines how differences in **business analytics (BA) resources, capabilities, and contextual factors** are associated with differences in firm performance.

## Empirical setting



- 125 firm-level studies
- 123 articles
- 2012–2021 research period
- 26 countries, four continents
- 358 effect sizes
- Quantitative synthesis through meta-analysis

# Meta-analysis results



| Independent variable       | n  | N      | Estimated effect | 95% CI      | Heterogeneity                                    |
|----------------------------|----|--------|------------------|-------------|--------------------------------------------------|
| BA technical resources     | 91 | 39,246 | 0.452            | 0.409–0.493 | Q = 1920.153**, df = 90, I <sup>2</sup> = 95.31% |
| BA human resources         | 42 | 9,219  | 0.494            | 0.444–0.541 | Q = 379.967**, df = 41, I <sup>2</sup> = 89.21%  |
| BA management capabilities | 69 | 16,074 | 0.477            | 0.433–0.519 | Q = 861.169**, df = 68, I <sup>2</sup> = 92.10%  |
| Data quality               | 14 | 3,088  | 0.434            | 0.328–0.528 | Q = 142.257**, df = 13, I <sup>2</sup> = 90.86%  |
| Organizational culture     | 18 | 3,698  | 0.472            | 0.391–0.546 | Q = 153.939**, df = 17, I <sup>2</sup> = 88.96%  |
| External pressure          | 17 | 3,644  | 0.165            | 0.036–0.288 | Q = 248.937**, df = 16, I <sup>2</sup> = 93.57%  |

## Key findings and implications

- BA is positively associated with firm performance across studies.
- The strongest effects are linked to **human resources, management capabilities, and organizational culture**.
- Technical resources and data quality matter, but they do not explain value creation alone.
- External pressure shows a smaller effect.
- High heterogeneity indicates that BA value creation remains context-dependent.



# Organizational perspectives



# A process perspective



Kunz et al. (2025): Process-level value creation from business analytics

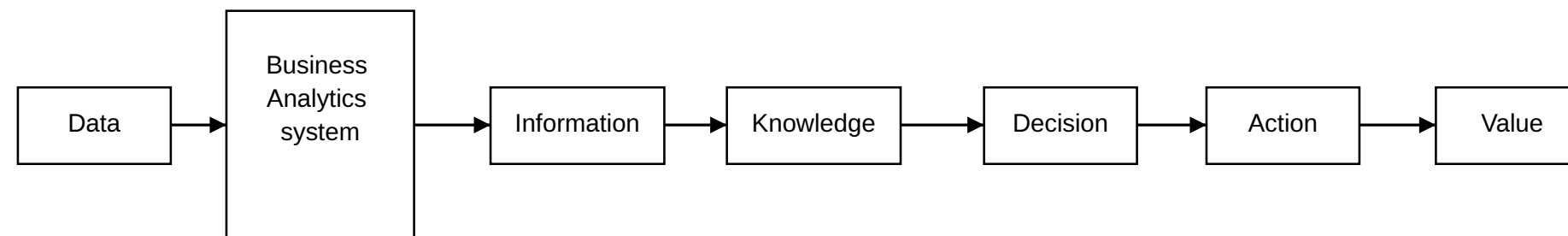
This theoretical literature review examines **how business analytics creates value in organizational processes** and how **machine learning changes these value-creation paths**.

## Research focus

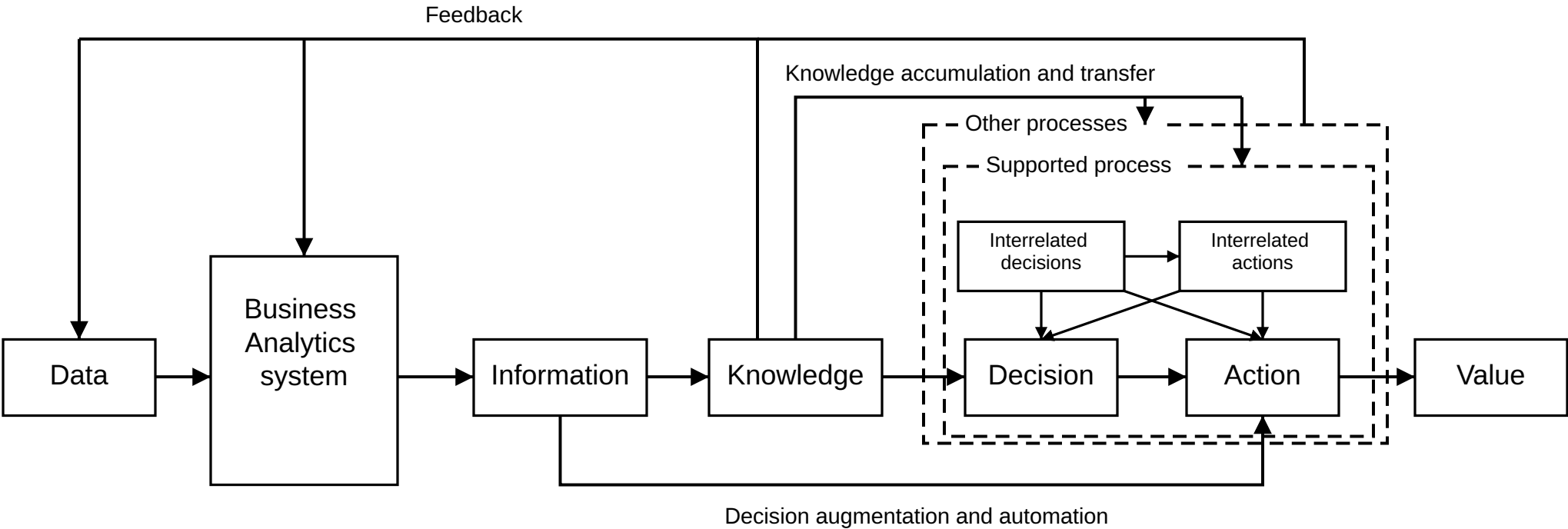
- Process-level value creation
- Business analytics systems in use
- Organizational decision-making
- Knowledge creation and use
- Effects of ML-based analytics

## Method and evidence base

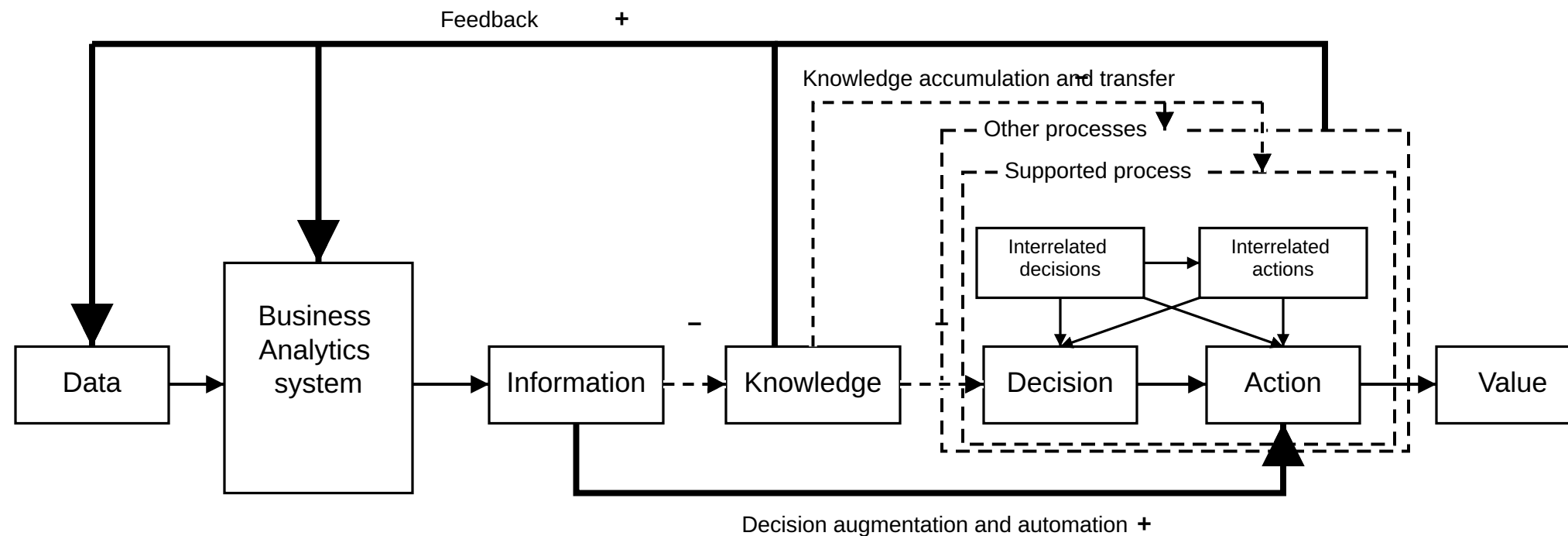
- Theoretical literature review
- Emergent theory building
- 176 articles synthesized
- Information Value Chain as sensitizing lens
- Six value-creation paths identified



# Value creation paths



# How machine learning changes value creation



- **Autonomy** leads to a stronger reliance on analytical information for decision-making and action
- **Dynamism** involves knowledge and data-driven feedback to improve the ground truth or the analytical system
- **Opacity** refers to the inscrutability of machine learning models, which restricts human understanding and process improvement

# Towards a mature analytics portfolio

The **TDWI Analytics Maturity Model** offers a diagnostic framework for assessing analytics capabilities across organization, resources, data infrastructure, analytics, and governance.

The model classifies analytics maturity into five stages:

- **Nascent → Early → Established → Mature → Advanced/Visionary**

TDWI’s model supports a structured assessment of where an organization’s analytics portfolio currently stands and where further development may be needed.

| Organizational Maturity                                                                                                 | Resource Maturity                                                                                                                           | Data Infrastructure Maturity                                                                                                                                                    | Analytics Maturity                                                                                                                                                             | Governance Maturity                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Leadership</li> <li>• Culture</li> <li>• Impact</li> <li>• Strategy</li> </ul> | <ul style="list-style-type: none"> <li>• Funding</li> <li>• Talent/skills</li> <li>• Roles/ responsibilities</li> <li>• Training</li> </ul> | <ul style="list-style-type: none"> <li>• Diversity, volume, and speed</li> <li>• Data access</li> <li>• Data integration and management</li> <li>• Data architecture</li> </ul> | <ul style="list-style-type: none"> <li>• Scope of capabilities</li> <li>• Automation/ augmented</li> <li>• Deployment and delivery approaches</li> <li>• Innovation</li> </ul> | <ul style="list-style-type: none"> <li>• Data governance processes and tooling</li> <li>• Model governance processes and tooling</li> <li>• Governance roles</li> <li>• Security/privacy</li> </ul> |

Based on Halper (2020)

# Responsible deployment: Analytics and AI risk management

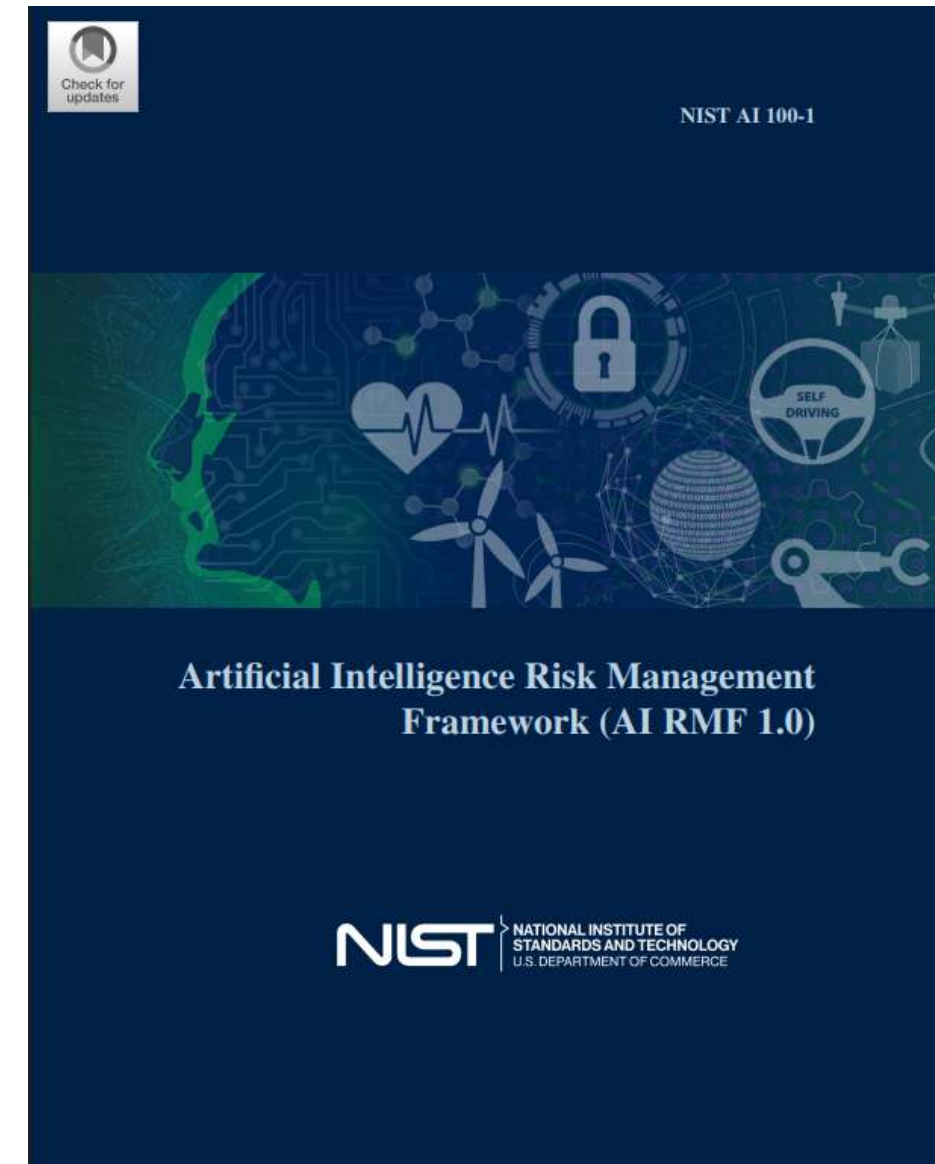
Advanced analytics increasingly becomes part of **AI systems** when it informs, recommends, or automates consequential decisions.

## NIST AI Risk Management Framework

- **Govern:** roles, policies, accountability
- **Map:** context, stakeholders, intended use
- **Measure:** performance, bias, robustness, risks
- **Manage:** mitigation, monitoring, incident response

## Deployment questions

- Who is affected?
- What decision is supported or automated?
- What can go wrong?
- Who is accountable?
- How do we know when to stop or intervene?



# Boundaries: what should not be optimized?



Analytics creates value only within legitimate organizational, ethical, and legal boundaries.

## Possible boundaries

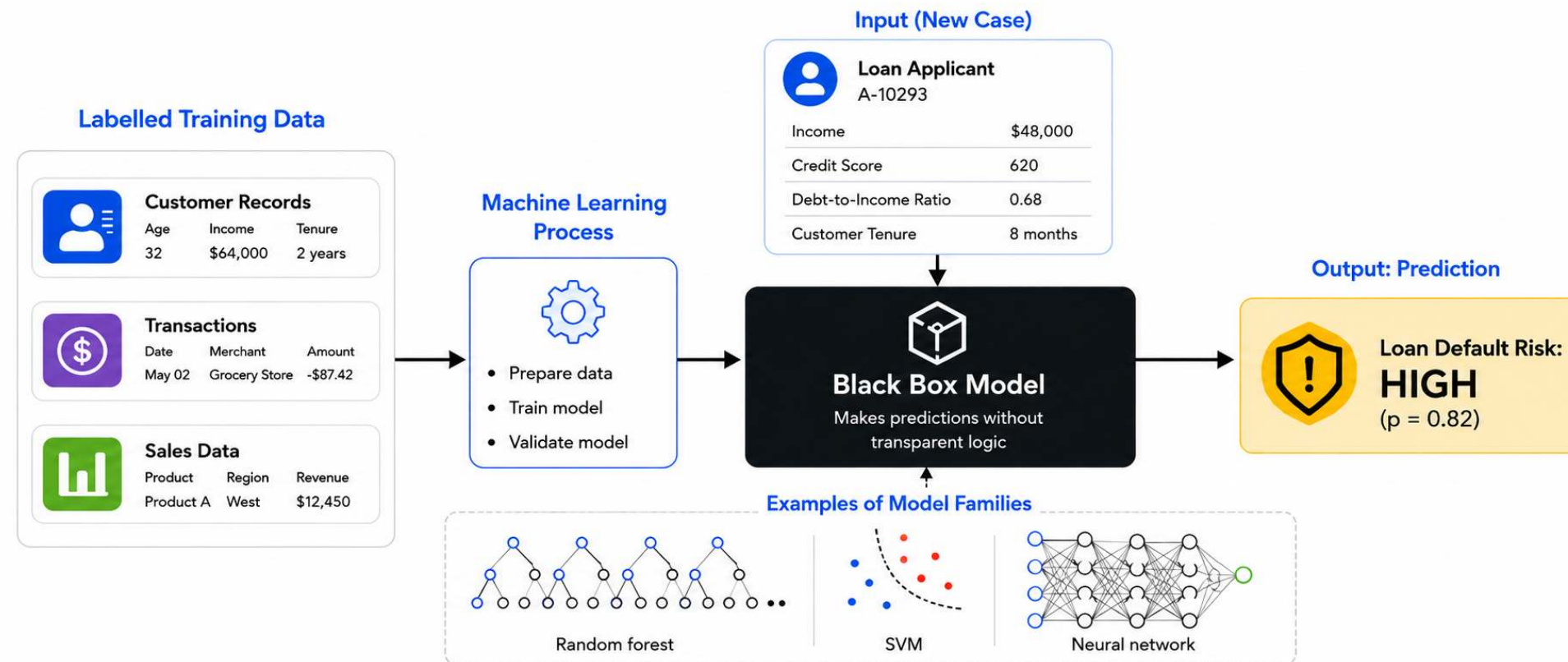
- Privacy and purpose limitation
- Non-discrimination and fairness
- Safety and reliability
- Human autonomy and dignity
- Legal compliance
- Organizational values

## Examples

| Use case           | Boundary question                     |
|--------------------|---------------------------------------|
| Employee analytics | Does this become surveillance?        |
| Credit scoring     | Who is disadvantaged?                 |
| Medical triage     | What happens in edge cases?           |
| Dynamic pricing    | When is personalization exploitative? |
| Fraud detection    | How are false positives handled?      |

# The black box of modern analytics

Modern analytics systems transform data into predictions without exposing a decision logic that people can easily inspect.

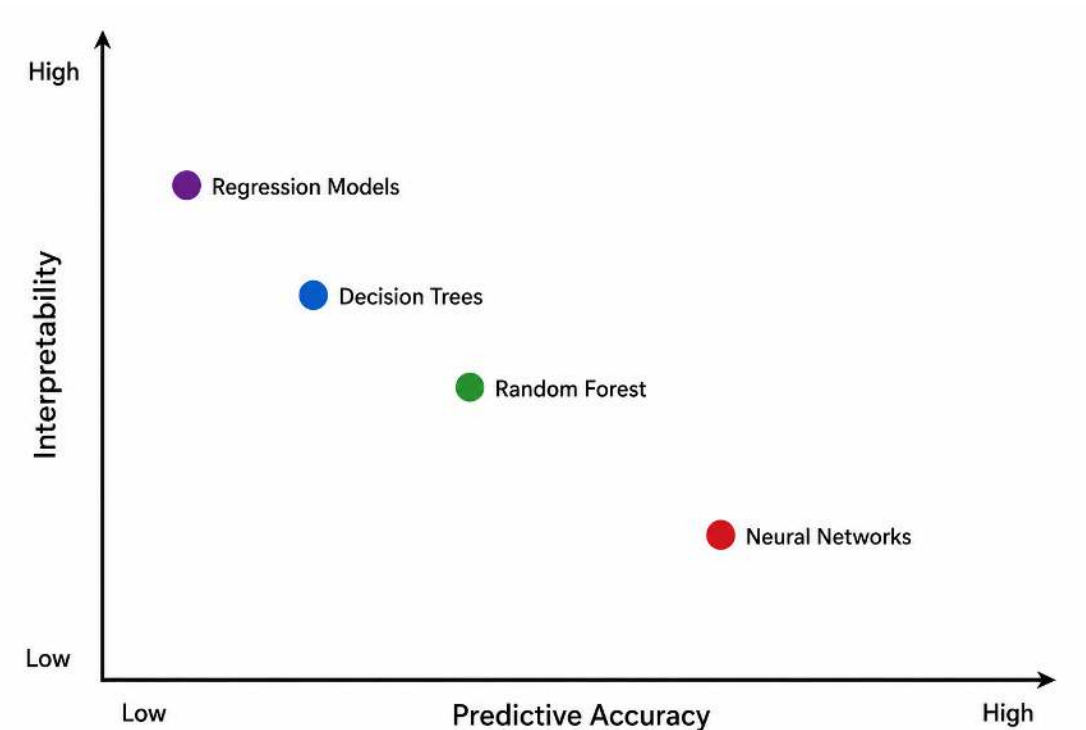


This raises practical questions:

- Why this prediction?
- Why not another prediction?
- When does the system fail?
- Who can contest or override it?



# The need for interpretability



Interpretability is needed for different purposes

| Context     | Purpose                                      |
|-------------|----------------------------------------------|
| Trust       | know when to rely on the output              |
| Engineering | debug errors, bias, and brittle behavior     |
| Governance  | assign responsibility and define oversight   |
| Recourse    | understand, challenge, or improve an outcome |
| Regulation  | document compliance and enable auditability  |

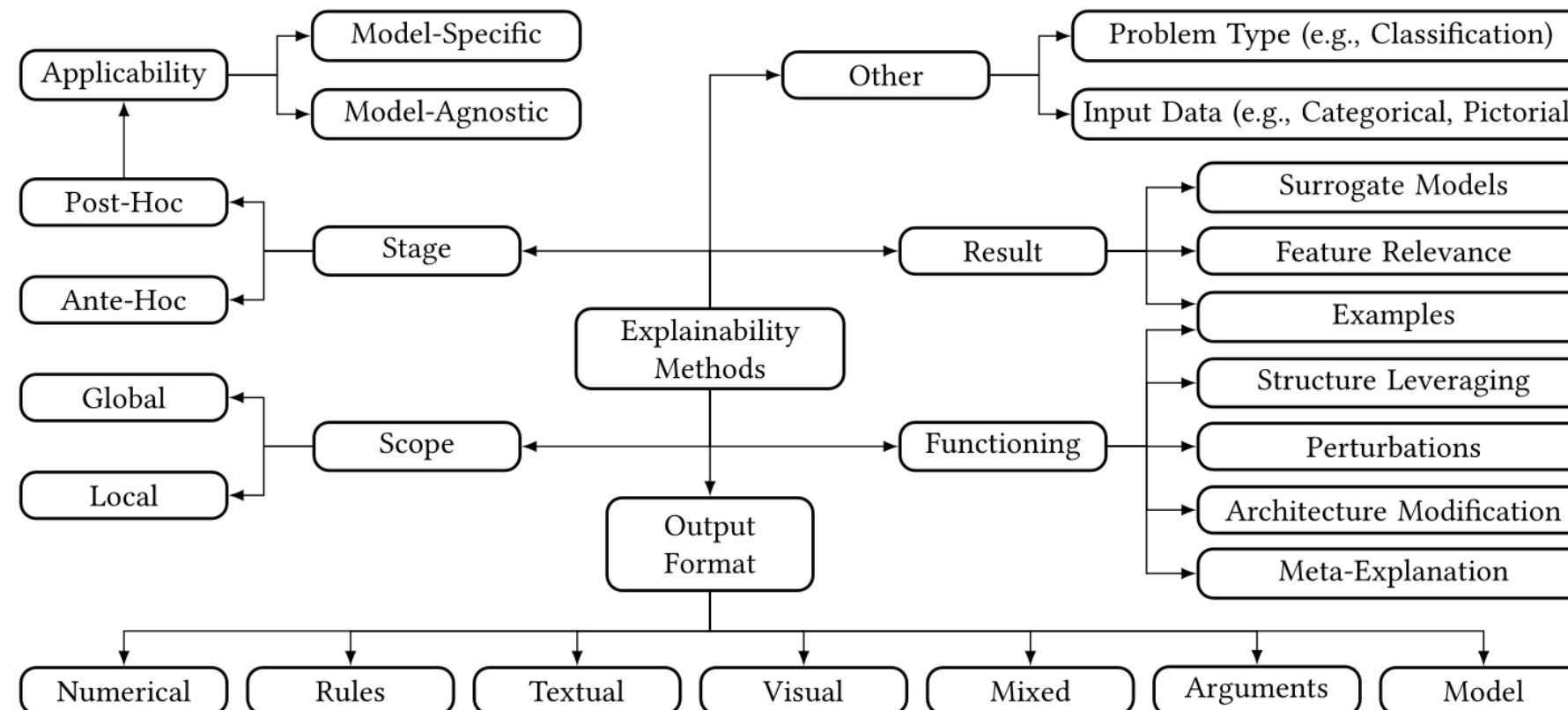
Based on Chamola et al. (2023)



# Explainability methods

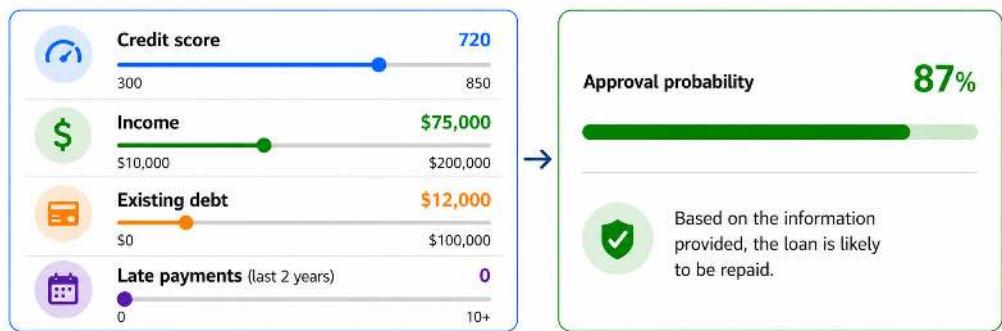
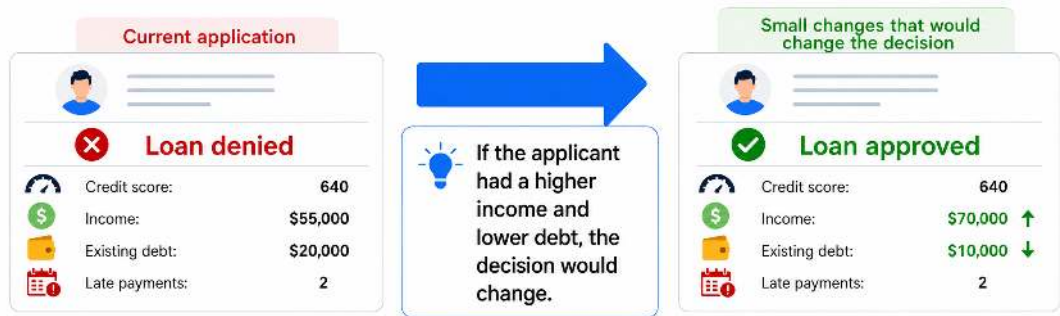


**Explainability methods** aim to make AI-based systems, their reasoning processes, or their outputs understandable, either by designing models that are explainable from the outset or by generating explanations for already-trained, potentially opaque models.



Based on Speith (2022)

# Explainability methods: Examples



## Feature importance

Mostly static analysis of the trained model and data; often post-hoc, though some models provide importance intrinsically. Usually provides a global view: which variables matter most across the model or dataset.

## Counterfactual explanations

Post-hoc explanation based on minimal feature changes that would alter the prediction.

## Interactive explanation

Post-hoc, user-driven exploration combining static explanations with what-if changes to input features.

# People perspectives

# Human resources for analytics

## BCG AI Radar 2025 survey

- 1,803 C-level executives
- Across markets, industries, roles, and company sizes

### Main finding:

AI is expected to reshape work more than eliminate workers.

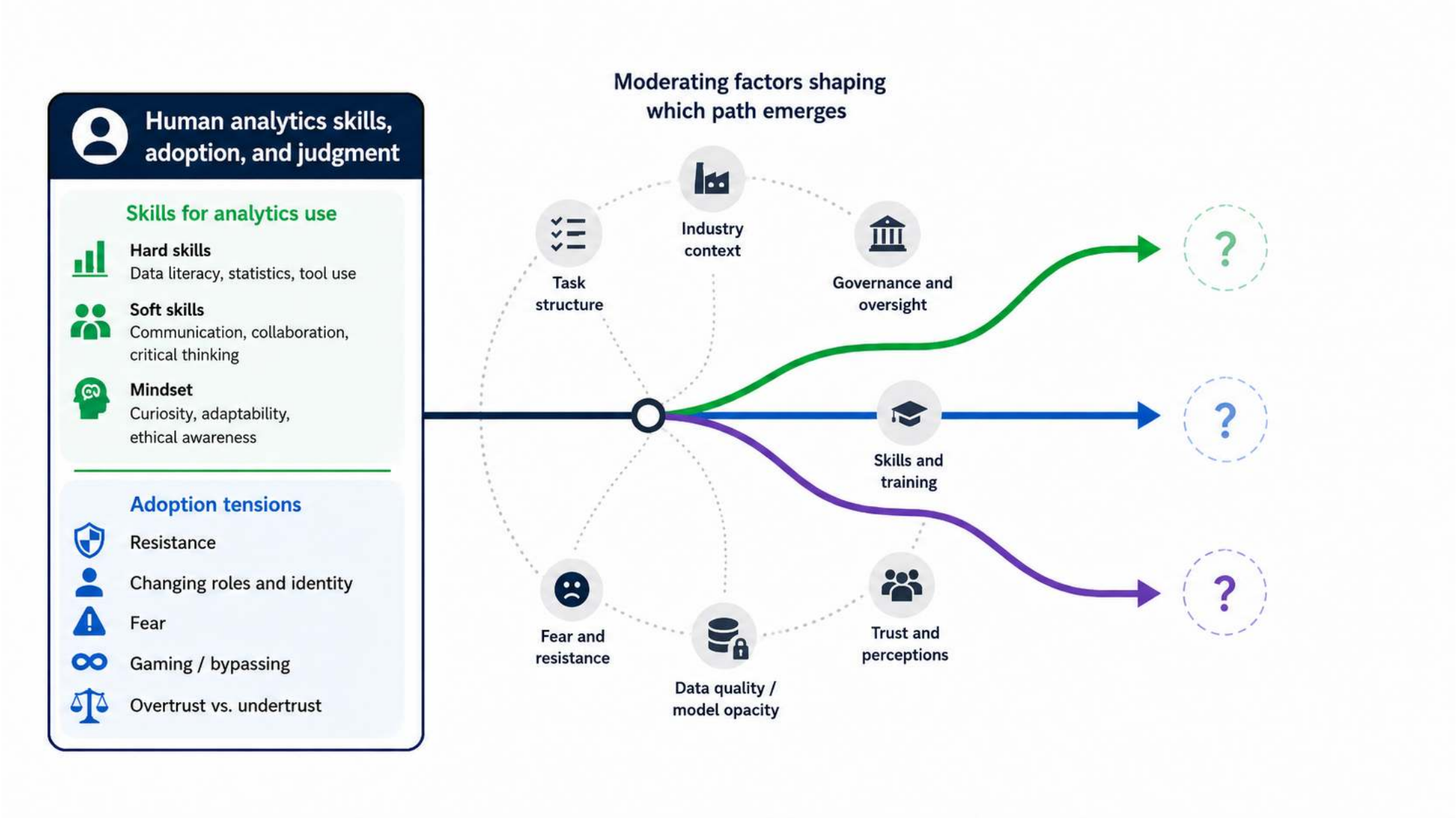
- Only 7% expect fewer FTEs due to automation
- 17% expect workforce restructuring with new roles
- 68% expect more productivity and upskilling of the existing workforce
- 8% expect more FTEs and new skills

Analytics and AI deployment is a **people-and-process transformation**: executives expect value to come from humans and AI working **side by side**, with human oversight remaining central.

Based on Apotheker et al. ([2025](#))



# Possible human futures in analytics and AI



# Summary



- Analytics can improve organizational performance, but its value depends on **context**, complementary resources, and how analytics is embedded in organizational processes.
- Value creation requires an analytics portfolio that combines **data infrastructure, technical and human resources, management capabilities**, and governance aligned with strategy.
- Responsible deployment means asking not only “Can we optimize this?” but also “Should we?” — defining ethical, legal, and organizational boundaries before systems go live.
- As analytics becomes more autonomous, dynamic, and opaque, organizations need interpretability, risk management, and oversight mechanisms that allow people to **understand, contest, and intervene**.
- The human role is changing rather than disappearing: analytics and AI create value when people have the skills, mindset, and organizational conditions to work effectively with them.

# Survey: Session 10



<https://forms.gle/Rhr4C47jsAZJyumN6>

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏



# References



- Apotheker, J., Duranton, S., Lukic, V., Bellefonds, N. de, Iyer, S., Bouffault, O., & Laubier, R. de. (2025). From potential to profit: Closing the AI impact gap: BCG AI radar. *Boston Consulting Group*. <https://www.bcg.com/publications/2025/closing-the-ai-impact-gap>
- Burrell, J. (2016). How the machine “thinks”: Understanding opacity in machine learning algorithms. *Big Data & Society*, 3(1), 2053951715622512. <https://doi.org/10.1177/205395171562251>
- Chamola, V., Hassija, V., Sulthana, A. R., Ghosh, D., Dhingra, D., & Sikdar, B. (2023). A review of trustworthy and explainable artificial intelligence (XAI). *IEEE Access*, 11, 78994–79015. <https://doi.org/10.1109/ACCESS.2023.3294569>
- Halper, F. (2020). *TDWI Analytics Maturity Model Assessment Guide*. TDWI Research. [https://go.tdwi.org/rs/626-EMC-557/images/TDWI\\_Analytics-Maturity-Model-Assessment-Guide\\_2020.pdf](https://go.tdwi.org/rs/626-EMC-557/images/TDWI_Analytics-Maturity-Model-Assessment-Guide_2020.pdf)
- Kunz, P. C., Spohrer, K., & Heinzl, A. (2025). Process-level value creation from business analytics: A theoretical literature review of value creation paths and changes induced by machine learning. *The Journal of Strategic Information Systems*, 34(2), 101902. <https://doi.org/10.1016/j.jsis.2025.101902>
- Müller, O., Fay, M., & Vom Brocke, J. (2018). The effect of big data and analytics on firm performance: An econometric analysis considering industry characteristics. *Journal of Management Information Systems*, 35(2), 488–509. <https://doi.org/10.1080/07421222.2018.1451955>
- Oesterreich, T. D., Anton, E., & Teuteberg, F. (2022). What translates big data into business value? A meta-analysis of the impacts of business analytics on firm performance. *Information & Management*, 59(6), 103685. <https://doi.org/10.1016/j.im.2022.103685>
- Speith, T. (2022). A review of taxonomies of explainable artificial intelligence (XAI) methods. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 2239–2250. <https://doi.org/10.1145/3531146.3534639>